



Himalayan Glacial Disaster — DRR Concept Diagram

Support options drawing on Japanese technology, following the 25–26 August 2026 Bhote Koshi–Trishuli flood in Nepal (working draft)

Drafted 2026-08-30 · Revised 2026-08-31 · Figures are NDRRMA as of 2026-08-31 09:00 NPT; search ongoing · Not an official position of any agency

The source mechanism has largely converged.
But the 6–10 hours downstream existed under every reading — and did not turn into evacuation.

1 What happened

	08:37	An ice mass falls on the north face of Langtang Lirung (~610 m wide, 0.2 km², 1,200 m drop). It briefly dams the Lhende , then breaches as one surge. M5.2 / Mw 5.7 came from the collapse
	08:50 09:20	Stage gauges stop transmitting (= destroyed). 09:05 DHM becomes aware, 16 min after the wave crossed the border
	09:15	Mass SMS alert: 679,295 messages . Only 4 districts
	11:45	A second collapse on the same slope (M4.2). Search teams are entering the valley — no warning issued
	15:20 16:00	Reaches Devghat → peak 6.57 m (no second peak)

About 10 hours from collapse to the lowest reach · 903 dead, 4,247 missing

This chronology moved four times in five days, and has now converged. First USGS report → IRDR (previous-day collapse + 18–19 h of impoundment) → satellite analysis (no dam) → **ICIMOD / USGS: a brief blockage and breach**. With no instruments upstream, everything rested on remote inference — which is what produced the swings. A GLOF struck the same corridor in July 2025 (11 dead).

2 Diagnosis — deaths concentrated where lead time was longest

District (upstream → downstream)	Margin after alert	Bodies recovered
Rasuwa	-25 min	24
Nuwakot	+5 min	95
Dhading	~1.5–2 h	55
Gorkha	~3–4 h	62
Tanahun	~4–5 h	37
Chitwan	6 h 05 min	272
Nawalparasi E not alerted	~7–8 h	207
Nawalparasi W not alerted	~8–9 h	151

A decisive caveat: “bodies recovered” is where a body was found, not where the person died, and recovery is rate-limited by regaining access (from 29 to 31 Aug the upstream counts caught up: Rasuwa 13 → 24, Nuwakot 51 → 95). That is exactly why **the verification survey (A-0) must precede any equipment donation**. The shape still holds: 457 people — **50.6% of the total** — were recovered in the four districts that were never alerted; the three lowest districts alone hold **69.8%**; Rasuwa, closest to the source, holds **2.7%**.

3 Four blind spots

- GAP 1 Pinning down the source took five days, four revisions**
USGS → IRDR (prolonged impoundment) → satellite (no dam) → **ICIMOD / USGS: a brief blockage and breach**. Readings diverge because there are no upstream instruments and everything is remote inference. That, in itself, is the case for putting monitoring there.
- GAP 2 Monitoring watches only for a rise in water**
The Rasuwagadhi gauge (446.22) showed no prior change. The upstream stations **stopped transmitting together after 08:40**, and that silence — itself the evidence of destruction — was logged as “missing data”. **The renewed rise on 30 Aug was likewise noticed by the river’s roar, not by an instrument**. Neither a falling stage nor missing data is treated as an anomaly.
- GAP 3 “A flood has occurred” does not move people**
670,000 messages went out at 09:15, yet the wave reached Chitwan **six hours later**. What was needed was not notice that something had happened but “what time it arrives at your village”, followed by tracking updates.
- GAP 4 Casualties concentrated among those SMS cannot reach**
Of the 4,247 missing, **592 are foreign nationals** (288 Indian, 90 American, 53 Ukrainian, 51 Malaysian, 35 Australian, 33 British and others), plus a large number of tunnel workers. Tunnels have neither signal nor escape routes; trekkers have neither Nepali-language SMS nor community networks. Neither group is in the design scope of the existing warning system.

4 The measures as a whole — laid out along the lead-time axis

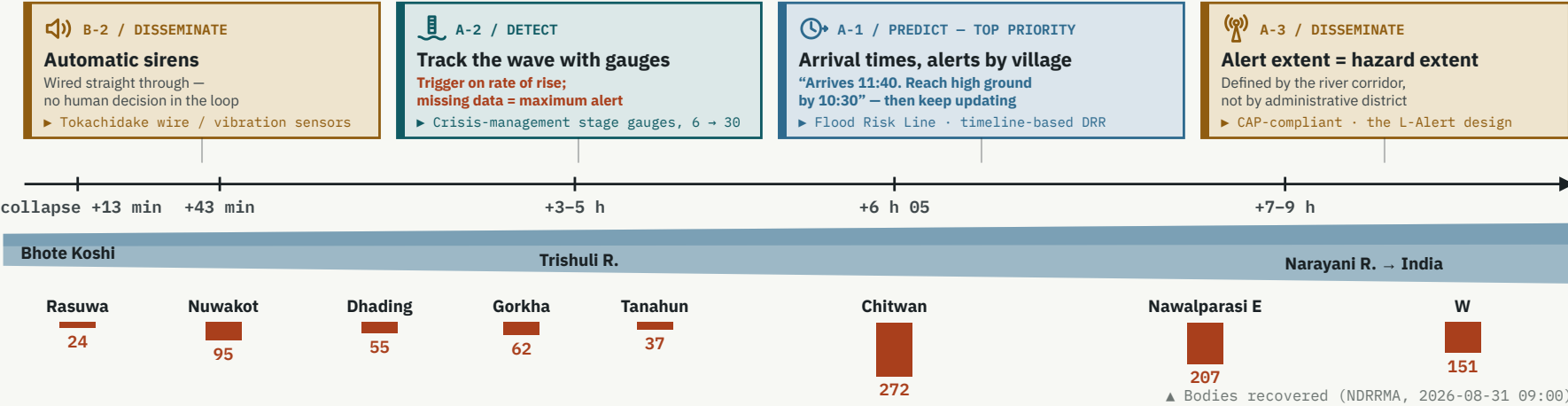
AT THE SOURCE — SETTLED AFTER FIVE DAYS

B-4 / TRANSBOUNDARY Peacetime monitoring Track change in lakes and in unstable slopes, continuously ► ALOS-2/4 · Sentinel Asia	B-1 / DETECT Collapse detection Alerts regardless of mechanism — so it holds under either hypothesis ► NIED Hi-net · microbarometers	B-3 / PREDICT — NEEDED HERE Blockage assessment It happened twice: in the initial mechanism, and again on 28 Aug ► Japan’s emergency-survey rules
---	--	---

SOURCE MECHANISM — FOUR REVISIONS IN FIVE DAYS, NOW CONVERGED

1 USGS: the 08:37 collapse is the origin → 2 IRDR: a 13:00 collapse the day before, breach after 18–19 h → 3 satellite: no dam 4 ICIMOD / USGS: the Langtang Lirung ice-rock avalanche briefly dams the Lhende, then breaches as one surge	
Orders of magnitude — only the excess constrains the source (now measured) Frictional melt (even spending all of the 1,200 m drop — 11.8 kJ/kg — caps melt at 3.5% of the ice mass) 10⁵–10⁶ m³ 18–19 h of channel blockage and impoundment (a catchment the size of the Lhende) 10⁶ m³ Total volume past Devghat (includes baseflow; cannot constrain the source). Peak flow 5,850 m³/s 10⁸ m³ ↳ the excess above baseflow — same order as the fallen ice (0.2 km²); friction can melt only 3.5% 2×10⁷ m³	
The renewed rise on 30 Aug — the same gap, reproduced four days later Flow rose again and NDRRMA alerted. Noticed not by an instrument but by a report that the river’s roar had grown . Search suspended.	the barrier lake was largely empty by 30 Aug

AFTER THE COLLAPSE — 100 KM IN UP TO 10 HOURS (THE ALERT WENT OUT; EVACUATION DID NOT FOLLOW)



Basin-wide, Peacetime	A-4 Protocols for tunnel workers and foreign nationals (cell broadcast / tunnel safety standards)	C-1 High-ground refuges and escape routes for every settlement	C-3 Hydropower plants as EWS nodes (monitoring obligations secured through loan and insurance conditions)	A-0 Verification survey (settle the source mechanism; census of where and when each death occurred and which alert was received)
------------------------------	--	---	--	---

Left: how the reading of the source moved four times in five days (as of 2026-08-31). Right: propagation of the flood wave after the collapse (the horizontal axis is not to scale). The margin at the source varies with the reading, but **the 6–10 hours downstream exist under every one of them**. Only three arrival times are measured — Syabrubesi 08:50, Betrawati 09:20, Devghat 15:20; the rest are interpolated. On the Trishuli, a 9 m rise at Galchchi and 7 m at Malekhu were recorded within 30 minutes. Bodies recovered is not a count by place of death, and recovery is rate-limited by regaining access.

5 Japanese technology and the bodies that would deliver it

DETECT NIED / Japan Meteorological Agency Hi-net, V-net; automatic seismic and infrasound detection of lahars Stations sit inside Nepal, so the system can be built without waiting for agreement with China . It needs no identification of mechanism, so it works while views still diverge.	DETECT MLIT / Instrument manufacturers Crisis-management stage gauges; wire and vibration sensors The “under ¥1 M, five years unpowered, retrofittable” specification adopted after Typhoon Lionrock in 2016 (24 dead in Iwaizumi).
PREDICT PWRI / ICHARM / MLIT Flood Risk Line; timeline-based DRR This disaster itself left measured data with which to calibrate the model .	DELIVER MIC / mobile network operators Emergency cell broadcast (ETWS / CB); L-Alert Broadcasts to every handset in several languages without any subscriber list, and reaches roaming foreign SIMs .
CROSS-BORDER JAXA / Sentinel Asia / ADRC ALOS-2 and -4 (L-band SAR); International Disasters Charter There is distinct value in Japan observing and distributing neutrally, as a third country .	STRUCTURAL MLIT Sabo Dept. / JICA / JBIC and NEXI Emergency survey of landslide dams; unmanned construction A statutory procedure that runs impounded volume → breach discharge → inundation estimate within 24–48 h . Needed both for the brief initial blockage and for the barrier lake that overtopped on 28 Aug.

6 Sequence of implementation

WHEN	WHAT	WHO
0–3 mo	A-0 verification survey — settle the blockage duration and the water budget; census of place and time of death and of which alert was received. In parallel, satellite monitoring of the unstable mass	Joint academic survey team / JAXA / ADRC
0–1 yr	A-1 arrival-time forecasting; A-3 review of alert coverage; A-4 tunnel and foreign-national protocols	ICHARM / JICA / MIC
1–2 yr	B-1 source detection network (alerts on mass movement regardless of mechanism); B-2 automatic sirens; B-3 transfer of the channel-blockage procedure. Stage gauges to 30+	NIED / MLIT Sabo Dept.
2–5 yr	C-1 high-ground refuges and escape routes; C-2 land-use guidance; C-3 power plants as nodes, with financial conditions	JICA / JBIC and NEXI
ongoing	Build the transboundary data-sharing framework; institutionalise Japan’s observation supply as a neutral third party	MOFA / ICIMOD / ADRC

On equipment dying within three years: require no mains power (five-year battery); use no consumables; duplicate the link (satellite IoT); hold spares and trained repairers locally; and pay against **95%+ annual uptime** rather than against delivery.

7 Stated for credibility: what this cannot do

- Satellites could not have predicted this ice–rock avalanche**
Brittle failure gives little precursory displacement. The role is limited to three things: **updating the lake inventory, rapid post-event assessment, and narrowing down unstable slopes**.
- Check dams cannot stop an event of this size**
Capturing a debris flow that travels over 100 km is not realistic. What works is high-ground refuges, training dykes on tributaries, and scour-resistant design of rebuilt structures.
- Precipitation products were powerless here**
No rain was involved in this event. The design must **give the rainfall and glacial paths independent detection chains and share only the downstream dissemination**.
- QZSS DC Report cannot be the primary path**
Nepal sits near the western edge of the service area, and deep valleys may not give sufficient elevation angle. **Start with a measured proof of concept on receivability**.

Legend (what the colours mean)

- Detection & observation Prediction & decision Dissemination & alerting Action & structures
 Cross-border & institutions Human losses

Group A = immediate (within 1 yr) / **Group B** = medium term (2 yr; where Japan is distinctively strong) / **Group C** = structures and land use (3–10 yr). Priority = lead time created ÷ cost × capacity to maintain.

Glossary

DHM: Dept. of Hydrology and Meteorology, Nepal
NDRRMA: National Disaster Risk Reduction and Management Authority
GLOF: glacial lake outburst flood
ICIMOD: Int’l Centre for Integrated Mountain Development
ICHARM: Int’l Centre for Water Hazard and Risk Management

CAP: Common Alerting Protocol
ETWS / CB: cell-broadcast emergency alerting
SAR: synthetic aperture radar (sees through cloud)
Unmanned construction: earthworks by remotely operated machines
Timeline DRR: planning backwards from the arrival time



Find it early upstream. Deliver time downstream.

What this disaster lacked was not a means of knowing the danger, but a path from knowing to escaping. On 30 Aug, too, it was a human ear that noticed. Sources: GFZ, ICIMOD, Copernicus / Planet satellite analysis, IRDR, USGS, NDRRMA, MLIT, NIED, JAXA, JICA and others